

Modeling Human Activity States Using Hidden Markov Models

Human Activity Recognition from Smartphone Accelerometer & Gyroscope Signals

Author: Nanen Miracle Mbanaade

Device & Sampling Rate: iPhone — 50 Hz

Accelerometer & gyroscope logged with the Sensor Logger app

[GitHub Repository](#)



1. Background and Motivation

Human Activity Recognition (HAR) turns raw inertial sensor streams into meaningful labels about what a person is doing. It underpins a wide range of applications, fitness and step tracking, context-aware mobile apps, rehabilitation monitoring, and elderly-care safety systems, because a modern smartphone already carries the accelerometer and gyroscope needed to sense motion continuously and unobtrusively. The core challenge is that the **true activity is never directly observed**; it must be inferred from noisy, high-rate sensor measurements. This makes a **Hidden Markov Model (HMM)** a natural fit: activities are hidden states that persist and transition over time in a structured way, while the sensor-derived feature vectors are the observable emissions. This project recognizes four activities; Still, standing, walking, and jumping from a single phone's accelerometer and gyroscope, and models their temporal dynamics with a Gaussian HMM.

2. Data Collection and Preprocessing

Data was collected with the Sensor Logger app on an iPhone, recording tri-axial accelerometer (acc_x/y/z) and gyroscope (gyro_x/y/z) at a 50 Hz sampling rate. Four activities were recorded: *standing, walking, jumping, and still* — each for 5–10 s per sample, with five recordings per activity for training and a separate held-out set reserved as unseen test data.

From raw export to model input

The Sensor Logger app exports each recording as a zip containing separate Accelerometer.csv and Gyroscope.csv files, each timestamped on a shared clock. A preprocessing script merges the two sensors for each recording into a single file with the six motion axes and a common timestamp. Because both sensors are sampled on the same clock at 50 Hz, the merge is a direct join with no interpolation required.

Windowing

Each recording is segmented into 1-second windows with 50% overlap. Each window becomes one observation vector for the HMM, preserving temporal order within a recording so that transition structure can be learned.

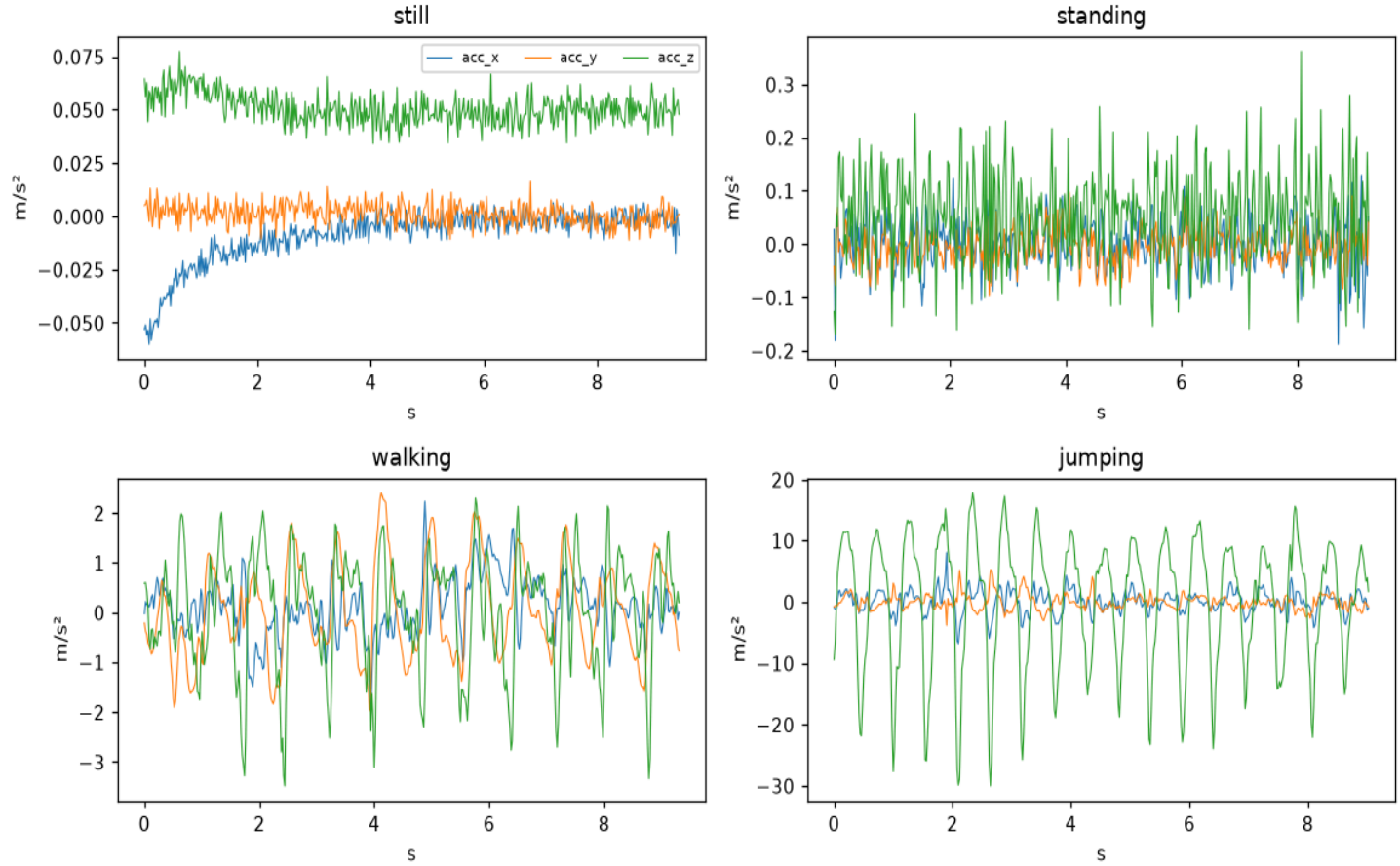


Figure 1. Representative raw accelerometer traces for each activity. Jumping shows large spikes; walking is periodic; still and standing are both low-amplitude, foreshadowing the difficulty of separating them.

3. HMM Setup and Implementation

The model is a four-state Gaussian HMM. The components map to the project specification as follows:

- Hidden states (Z): still, standing, walking, jumping.
- Observations (X): per-window feature vectors (26 features).
- Transition matrix (A): learned $P(\text{activity}_t \rightarrow \text{activity}_{t+1})$.
- Emission model (B): per-state multivariate Gaussian (diagonal covariance) over standardized features.
- Initial distribution (π): learned probability of the starting activity.

Features

Time-domain: per-axis mean, standard deviation, and variance; signal magnitude area (SMA); resultant acceleration magnitude mean/std; and inter-axis correlations (xy, xz, yz). Frequency-domain (from the FFT of the acceleration magnitude): dominant frequency and spectral energy. These capture both the intensity and the periodicity that distinguish the activities.

Algorithms

Baum–Welch (Expectation–Maximization) estimates A, B and π from the observation sequences; training converged in a handful of iterations. **Viterbi** decoding then recovers the single most likely hidden-state sequence. A from-scratch NumPy Viterbi is included in the notebook and verified to produce the same

decoding as the library implementation. Because HMM state indices are arbitrary, learned states are aligned to ground-truth activity labels by the best-matching permutation before scoring.

4. Results and Interpretation

The learned transition matrix (Figure 2) is strongly diagonal-dominant: each activity is most likely to be followed by itself in the next window. This is behaviorally realistic; a person does not change activity every fraction of a second and shows the HMM captured temporal persistence rather than treating windows independently.

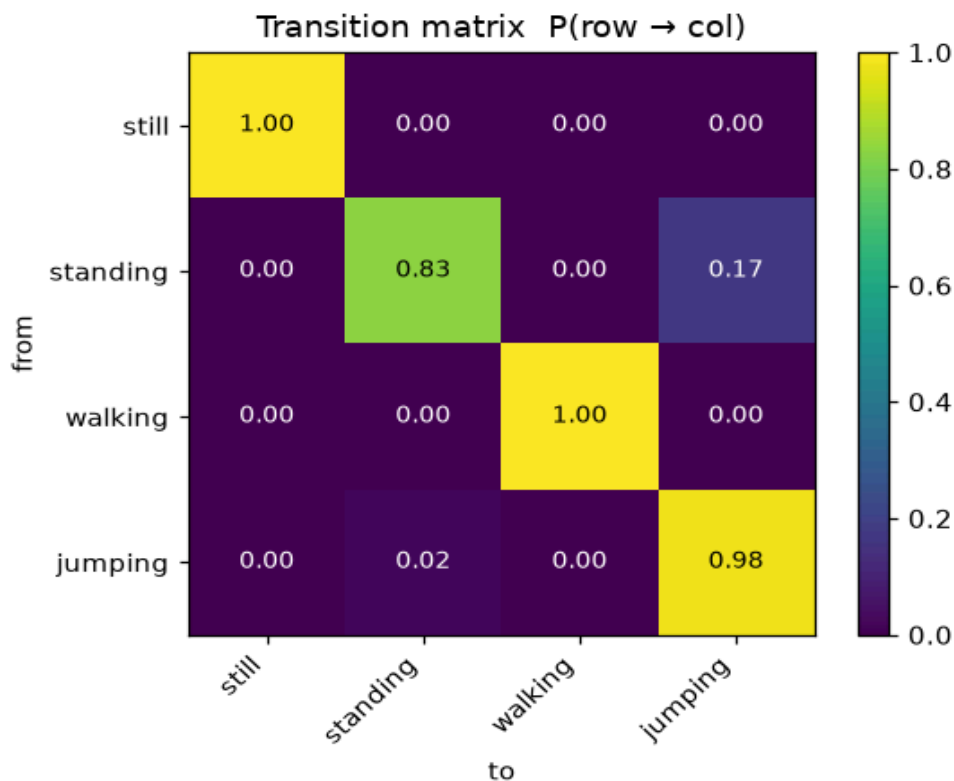


Figure 2. Learned transition matrix $P(\text{row} \rightarrow \text{col})$. High diagonal values indicate activities persist across adjacent windows.

Viterbi decoding tracks the ground-truth sequence for the high-motion activities closely (Figure 3).

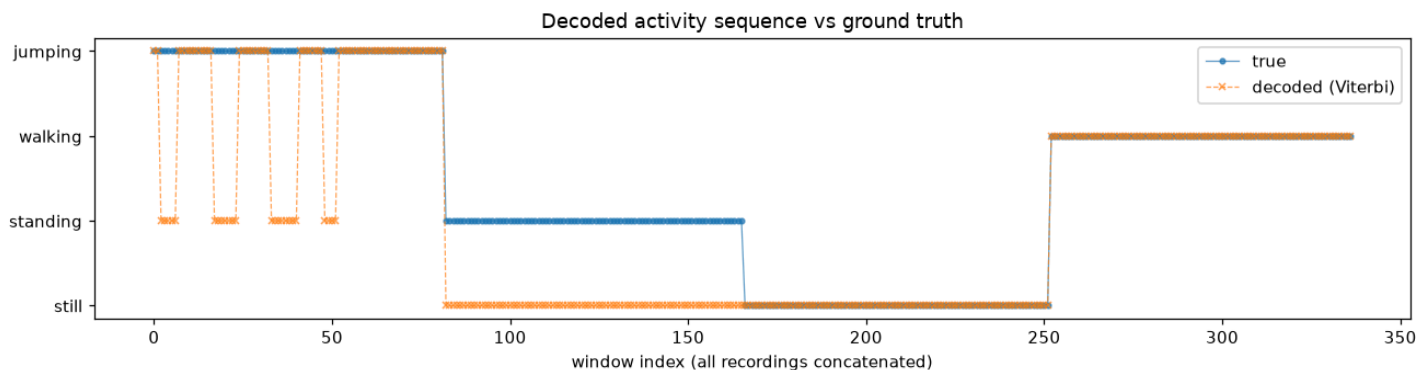


Figure 3. Decoded (Viterbi) activity sequence overlaid on ground truth across all concatenated recordings.

Evaluation on unseen data

The trained model was evaluated on a separate recording session not used in training. Features were standardized with the training statistics, and states were mapped with the training permutation, so no information leaked from test to training. Per-class results:

State (Activity)	Number of Samples	Sensitivity	Specificity	Overall Accuracy
Still	49	1.00	0.89	0.92
Standing	50	0.00	1.00	0.75
Walking	51	1.00	0.77	0.83
Jumping	49	1.00	1.00	1.00

Overall unseen accuracy: ~0.75. The aggregate figure conceals a highly uneven per-class picture, which is the most informative part of the result.

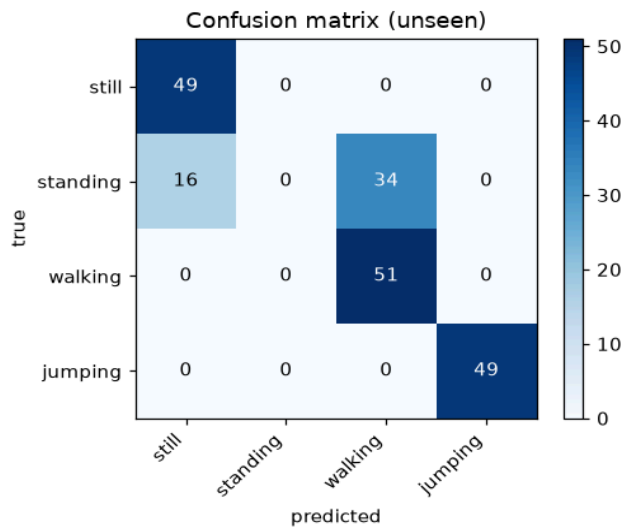


Figure 4. Confusion matrix on the held-out unseen session. The standing row is absorbed into the still column.

5. Discussion and Conclusion

Which activities were easiest and hardest

Jumping was recognized perfectly (sensitivity and specificity 1.00): its large vertical acceleration spikes and high spectral energy make it unmistakable. It was also recognized very well (sensitivity 1.00, ~92% accuracy), since the phone resting on a hard surface produces an almost mechanically flat signal. Walking was recognized reliably (sensitivity 1.00, ~83% accuracy) thanks to its clear periodic stepping rhythm. Standing was the clear failure case, with a sensitivity of 0.00 — not a single standing window was labelled correctly, as they were all absorbed into the still state.

Why standing failed

This confusion was anticipated before data collection. When the phone is gripped firmly while standing, its motion signal is only marginally more active than when it lies still on a surface. Once the raw signal is compressed into window-level features, the two low-motion classes become nearly indistinguishable to the Gaussian emissions. The specificity of 1.00 for standing confirms the direction of the error: the model never assigned other activities to standing; instead, standing itself was consistently mistaken for still.

Transition probabilities and realism

The diagonal-dominant transition matrix reflects genuine behavior activities that are sustained over time rather than switching every window, which indicates the temporal model is doing meaningful work beyond per-window classification.

Effect of noise and sampling rate

The 50 Hz rate was more than sufficient to resolve the periodic structure of walking and jumping. The limiting factor was not sampling rate but the intrinsic similarity of the two low-motion classes: small sensor noise dominates the still/standing distinction, which is why that pair collapsed while the high-energy classes remained cleanly separated.

Improvements

- Record standing with natural, relaxed body sway rather than a firm, frozen grip, so it is physically more distinct from still.
- Add features emphasizing low-frequency motion and postural sway to separate the two low-motion classes.
- Collect more recordings per activity to improve the Gaussian emission estimates.
- Add a magnetometer or orientation feature for additional context.

Conclusion

The Gaussian HMM successfully recovered interpretable activity states from raw smartphone inertial data, recognizing jumping, still, and walking reliably on unseen data and learning realistic, persistent transition dynamics. Its main limitation is the still versus standing distinction: because both are low-motion states, standing was systematically absorbed into still, pulling overall unseen accuracy to roughly 75%. This is a physically meaningful outcome rather than a defect; the model's discriminative power is bounded by how distinct the underlying signals actually are. The most promising next step is to make standing physically more distinct during collection and to add low-frequency motion features, targeting exactly the pair that the current model confuses.